Vodafone Proactive Retention Engine

From Predictive Insight to Autonomous Retention Workflows

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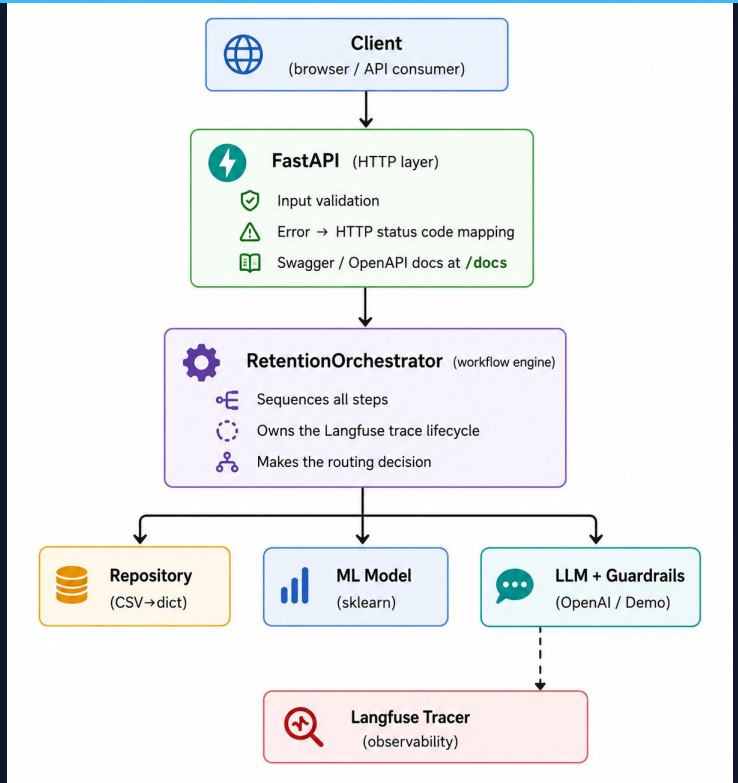
DEPT® AI Engineer Case Assignment - May 2025

Why Existing Churn Models Are Not Enough?

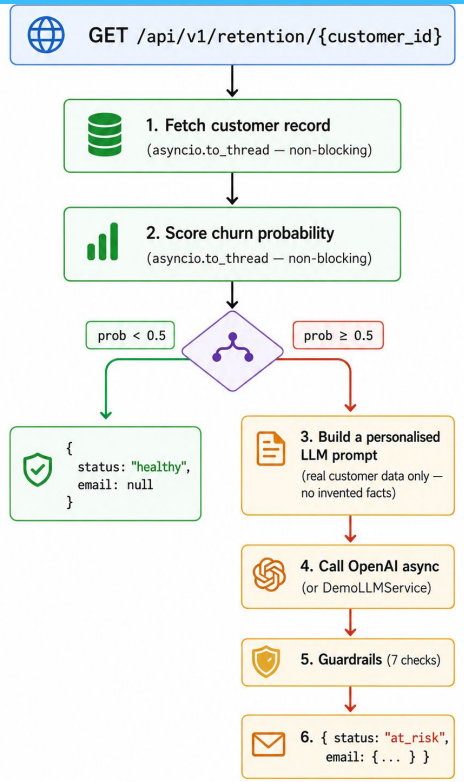
- Predictive models alone do not reduce churne Operationalization Gap
- Need automated, scalable retention workflows
- Manual personalization does not scale
- Brand-safe AI generation is critical

Production-Ready Retention Engine



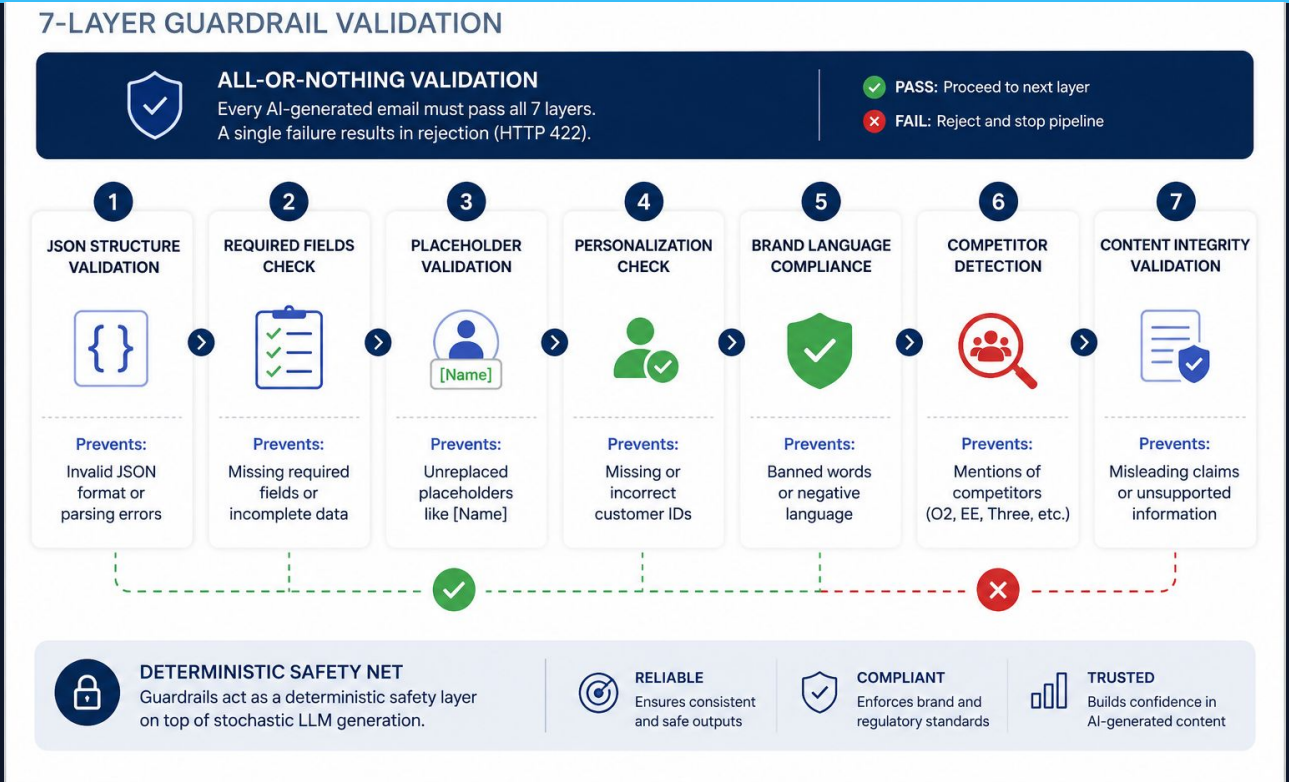
- Production-oriented modular architecture with separated orchestration, inference, generation, and validation layers to ensure scalability, maintainability, reliability, and enterprise-ready AI delivery.

End-to-End AI Workflow



- The workflow transforms churn predictions into automated retention actions by combining customer retrieval, ML inference, intelligent routing, GenAI generation, and guardrail validation within a scalable orchestration pipeline.

Enterprise Safety Pipeline



Scalability Strategy

ASYNC FASTAPI
Built with FastAPI for high-performance asynchronous handling

NON-BLOCKING INFERENCE
Async I/O and concurrent tasks ensure efficient resource utilization

STATELESS SERVICES
Stateless design enables horizontal scaling and high availability

STARTUP OPTIMIZATION
Heavy models & resources are loaded at startup for faster request handling

CURRENT OPTIMIZATIONS

Async Endpoints
Full async/await implementation for all I/O operations

Non-Blocking Inference
ML inference and LLM calls run without blocking the event loop

Startup Initialization
Models, clients, and resources are loaded once at startup

Stateless API Design
No in-memory session/state, safe to scale horizontally

Efficient Resource Usage
Connection pooling, reuse of clients, and optimized dependencies

PRODUCTION ARCHITECTURE (CURRENT)

Clients → FastAPI (Async) → Retention Orchestrator → Churn Prediction Service / LLM Generation Service / Guardrails Validation → Observability (Langfuse) → (Future) PostgreSQL

SCALABILITY ROADMAP

REDIS CACHE
Cache frequent data & responses to reduce latency and external calls

→

QUEUE LAYER
Use message queues (e.g., RabbitMQ) for background tasks and high throughput

→

POSTGRESQL
Migrate from SQLite to PostgreSQL for ACID compliance and scalability

→

KUBERNETES
Container orchestration for auto-scaling, self-healing and rolling deployments

INFRASTRUCTURE & SCALING

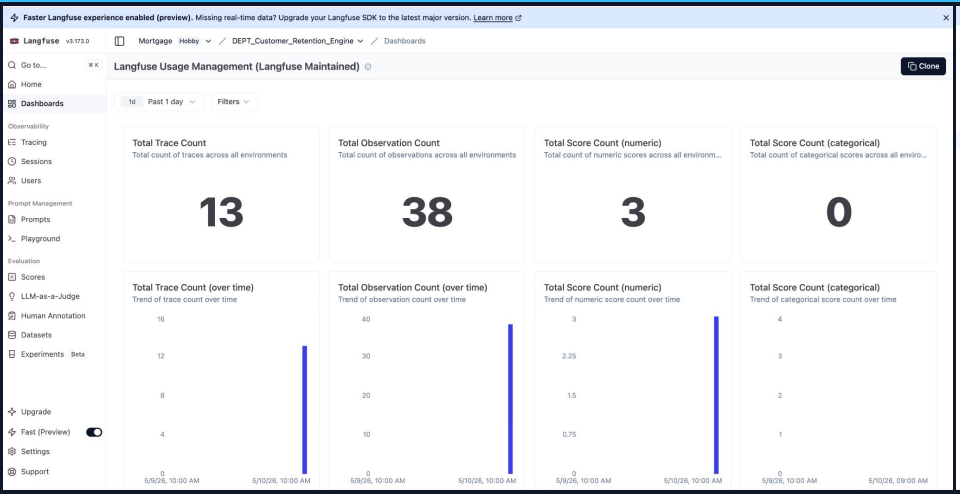
DOCKERIZED SERVICES
Consistent environments and easy deployment

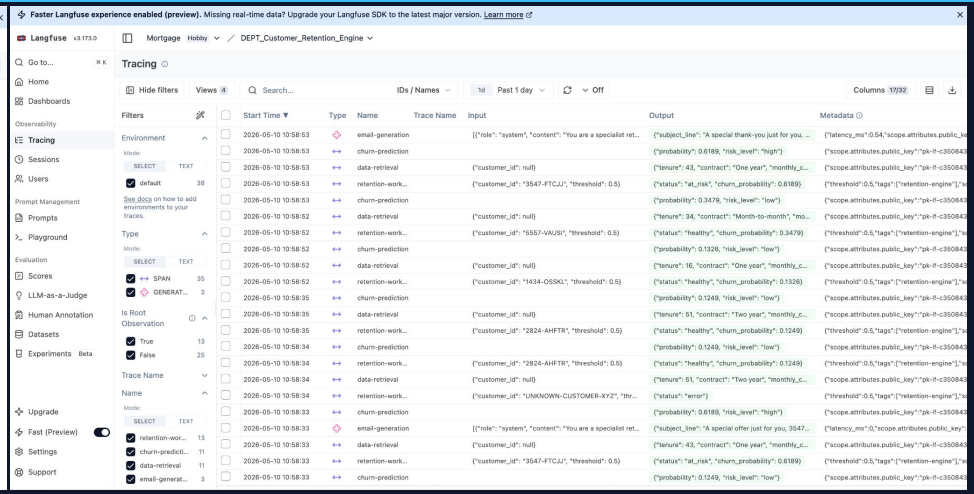
HORIZONTAL SCALING
Add more instances behind load balancer

HIGH AVAILABILITY
Health checks, retries and resilient design

CLOUD READY
Ready for deployment on AWS / GCP / Azure

Production Observability





Production observability is implemented to make the AI workflow traceable, measurable, and debuggable. Langfuse captures LLM prompts, responses, latency, token usage, and generation events, while structured logs and monitoring metrics help track API performance, guardrail failures, routing decisions, and overall system reliability in a production environment.

API Demo

127.0.0.1:8000/docs

Vodafone Proactive Retention Engine 1.0.0 OAS 3.1

/openapi.json

A modular AI service that operationalises the Vodafone churn prediction model into a fully autonomous retention workflow.

How it works

1. **Fetch** — retrieve the customer's profile from the database.
2. **Score** — run the trained sklearn model to get a churn probability.
3. **Route**:
   - `churn_probability < 0.5` → return `status: healthy` immediately (no LLM cost).
   - `churn_probability ≥ 0.5` → generate a personalised retention email via OpenAI, validated through a 7-check brand guardrail before returning.

Demo mode

No OpenAI API key? The service runs in **demo mode** automatically — template-based emails are generated and every other pipeline stage (ML inference, guardrails, Langfuse tracing) runs fully.

Alternatively, pass your own key per-request via the `X-OpenAI-Key` header.

Example customer IDs to try

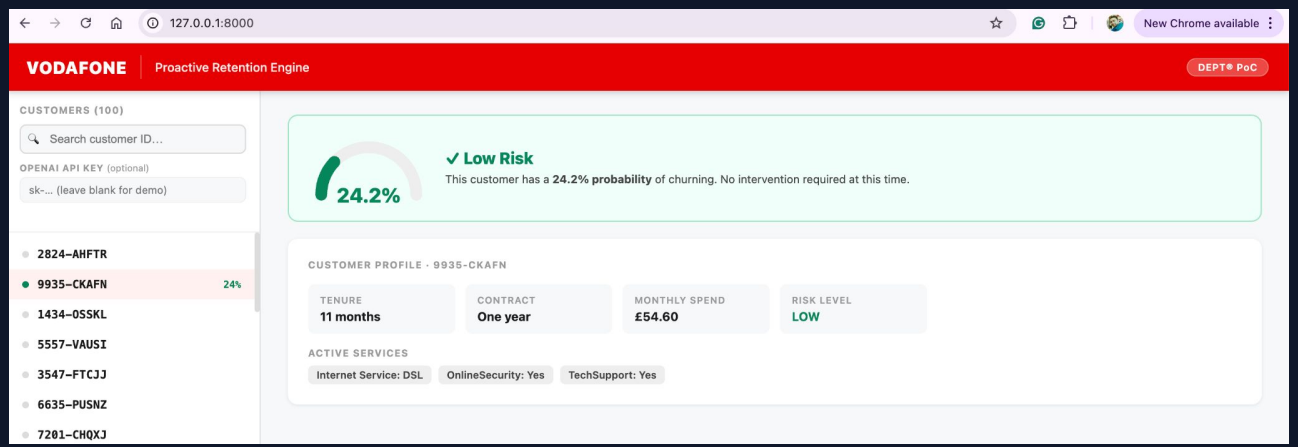
Customer ID	Expected result
`1053-YWGNE`	High-risk → retention email generated
`3170-YWWJE`	Low-risk → healthy, no email

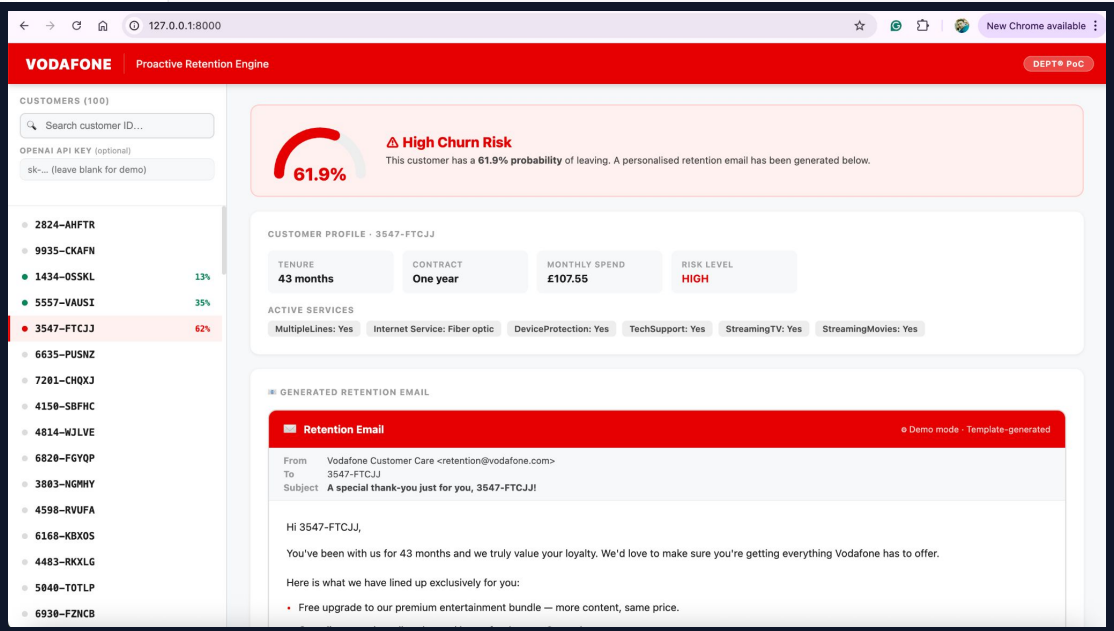
Source code

github.com/soheil2017/Customer_churn_DEPT

Private — Case Assignment

Frontend Demo





Why This Architecture?

MODULAR & MAINTAINABLE
Clear separation of concerns across services makes the system easy to maintain, test, and evolve.

ORCHESTRATION DRIVEN
Central orchestrator manages the end-to-end workflow, enabling intelligent routing, retries, and error handling.

RELIABILITY BY DESIGN
7-layer guardrails ensure safe, brand-compliant, and deterministic outputs on top of stochastic LLM generation.

SCALABILITY FIRST
Stateless services, async processing, and queue-based architecture enable horizontal scaling and high throughput.

OBSERVABILITY BUILT-IN
End-to-end tracing, metrics, and logging provide full visibility into prompts, responses, latency, and system health.

BUSINESS IMPACT FOCUSED
Transforms churn predictions into personalized, automated retention actions that drive real business outcomes.

KEY OUTCOMES

Enterprise-grade AI system
Safe, reliable, and production-ready.

Scalable & Future-Ready
Designed to handle growth and evolving requirements.

Operational Efficiency
Automated workflows reduce manual effort and time-to-action.

Customer-Centric
Personalized, relevant communication improves engagement and retention.

FUTURE ROADMAP

PostgreSQL
Migrate from SQLite for robust data management

→

Redis Caching
Improve latency and reduce repeated computations

→

CRM Integration
Seamless integration with enterprise CRM platforms

→

Human-in-the-Loop
Approval workflows for high-impact retention actions

Key Takeaways

Reliability

Production

Business Value

Built to operationalize AI safely at scale

Thank You